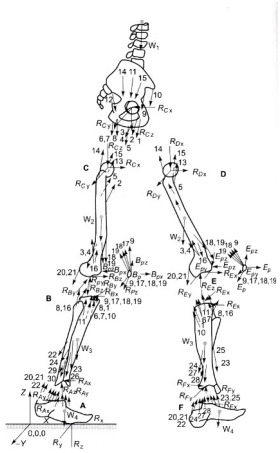
Lecture 4: Dynamic analysis of movement

SM2501/FSG2084



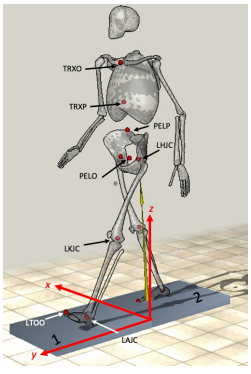
Robertson et al 2004

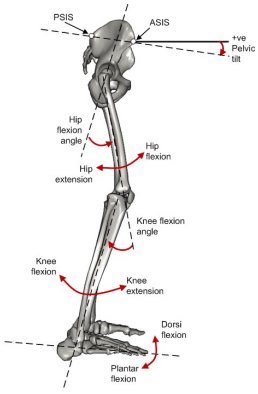
Objectives:

- Introduction of kinetics analyses
- Theory and application of inverse dynamics
- Theory of 3D rigid body kinematics

1

But first… comments on computation of segment/joint kinematics





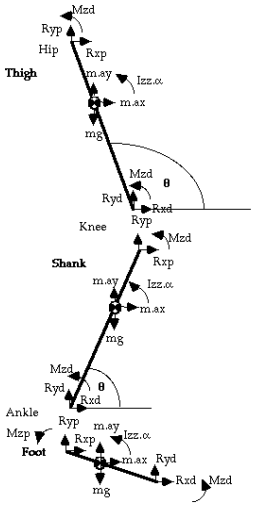
Trunk: absolute angle
Pelvis: absolute angle
Hip: femur relative to pelvis
Knee: tibia relative to femur
Ankle: foot relative to tibia

***Note**: report angles in degree (but compute derivatives in radians)*

*Remember to do **sanity checks** on your results.*

2

Kinetics and inverse dynamics



Why?

Problem: We can measure motion (kinematics) and external forces (GRF), but we cannot directly measure internal joint moments

Question: Given a representative model of a subject, joint kinematics over time, and measurements of the external forces applied to the subject, **find the net joint forces and moments** that must have been present to produce the given motion.

Approach: Work backward from motion + external forces → calculate joint moments, i.e. NET effect of all muscles and structures crossing the joint.

”Inverse" - Effect → Cause

3

What is inverse dynamics?

INPUT:

✓ Body segment model (mass, inertia, dimensions)

✓ Joint kinematics (angles, velocities, accelerations)

✓ External forces (ground reaction forces)

PROCESS:

→ Apply Newton's laws to each segment

→ Work from ground upward (distal to proximal)

OUTPUT:

✓ Net joint reaction forces

✓ Net joint reaction moments

✓ Joint powers

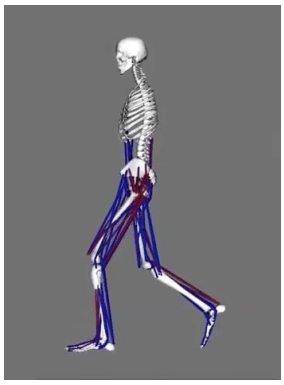
4

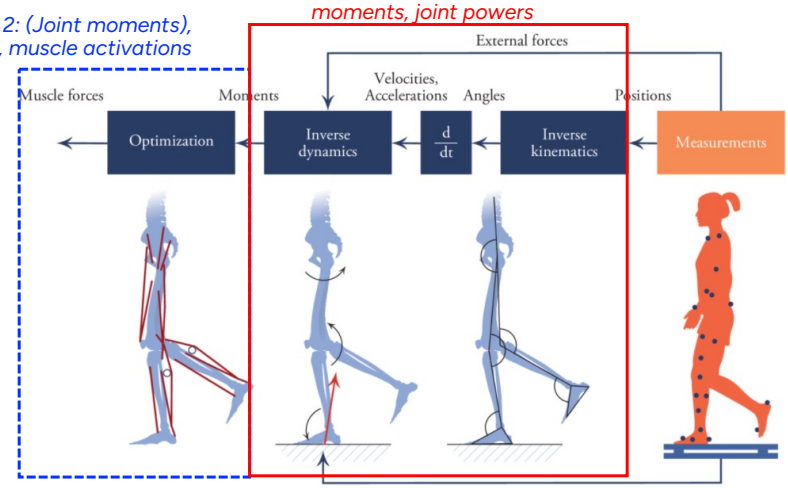
5

Inverse dynamics analysis

Project 2: (Joint moments), muscle forces, muscle activations

Project 1: Angles, joint moments, joint powers





Uchida and Delp 2020

5

Newton's laws: relationship between force and linear motion of particles or point masses

1. *Every object perseveres in its state of rest, or of uniform motion in a right line, except insofar as it is compelled to change that state by forces impressed thereon*
   - *Principle of inertia*
2. *The change of motion of an object is proportional to the force impressed; and is made in the direction of the straight line in which the force is impressed.*

   For particles or point masses:

   $\mathbf{p} = m\mathbf{v}$

   $\mathbf{F} = m\mathbf{a}$
3. *To every action, there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.*

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Euler's laws: dynamics of rigid bodies

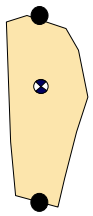
1. The rate of change of linear momentum **p** of a rigid body with mass *m* is equal to the resultant of all the external forces $\mathbf{F}_{ext}$ acting on the body
   $$\sum \mathbf{F}_{ext} = \frac{d\mathbf{p}}{dt} = m\mathbf{a}_{COM}$$

2. The rate of change of angular momentum **H** about a point *p* that is fixed in an inertial reference frame, often the center of mass (COM) of the body, is equal to the sum of the external moments/torques **M** acting on that body about that point *p*:
   $$\sum \mathbf{M}_p = \frac{d\mathbf{H}_p}{dt} = \sum \mathbf{r}_{pA} \times \mathbf{F} + \mathbf{M}_{ext} = I_p \boldsymbol{\alpha}$$

   Note: point *p* can be the body COM or a fixed point, but *not* any random point.

7

Anthropometry: segment lengths, masses and inertial characteristics



- Physical measurements of body with respect to different individuals
  - E.g. segment mass as % of body mass
  - Segment COM locations, e.g. thigh
  - Moment of inertia I for segment *j* is generally described as radius of gyration ρ_j wherein $I_{Gj} = m_j \rho_{Gj}^2$
- Segment masses often based on densities and volumes, with separate densities for bone, muscle, fat, lungs and water
- Segment COMs based mostly on cadaver studies (balancing segments)

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Anthropometry table, example Winter 2009

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TABLE 4.1 Anthropometric Data

Segment	Definition	Segment Weight/Total Body Weight	Center of Mass/ Segment Length Proximal	Center of Mass/ Segment Length Distal	Radius of Gyration/ Segment Length C of G	Radius of Gyration/ Segment Length Proximal	Radius of Gyration/ Segment Length Distal	Density
Hand	Wrist axis/knuckle II middle finger	0.006 M	0.506	0.494 P	0.297	0.587	0.577 M	1.16
Forearm	Elbow axis/ulnar styloid	0.016 M	0.430	0.570 P	0.303	0.526	0.647 M	1.13
Upper arm	Glenohumeral axis/elbow axis	0.028 M	0.436	0.564 P	0.322	0.542	0.645 M	1.07
Forearm and hand	Elbow axis/ulnar styloid	0.022 M	0.682	0.318 P	0.468	0.827	0.565 P	1.14
Total arm	Glenohumeral joint/ulnar styloid	0.050 M	0.530	0.470 P	0.368	0.645	0.596 P	1.11
Foot	Lateral malleolus/head metatarsal II	0.0145 M	0.50	0.50 P	0.475	0.690	0.690 P	1.10
Leg	Femoral condyles/medial malleolus	0.0465 M	0.433	0.567 P	0.302	0.528	0.643 M	1.09
Thigh	Greater trochanter/femoral condyles	0.100 M	0.433	0.567 P	0.323	0.540	0.653 M	1.05
Foot and leg	Femoral condyles/medial malleolus	0.061 M	0.606	0.394 P	0.416	0.735	0.572 P	1.09
Total leg	Greater trochanter/medial malleolus	0.161 M	0.447	0.553 P	0.326	0.560	0.650 P	1.06
Head and neck	C7–T1 and 1st rib/ear canal	0.081 M	1.000	— PC	0.495	0.116	— PC	1.11
Shoulder mass	Sternoclavicular joint/glenohumeral axis	—	0.712	0.288	—	—	—	1.04
Thorax	C7–T1/T12–L1 and diaphragm*	0.216 PC	0.82	0.18	—	—	—	0.92
Abdomen	T12–L1/L4–L5*	0.139 LC	0.44	0.56	—	—	—	—
Pelvis	L4–L5/greater trochanter*	0.142 LC	0.105	0.895	—	—	—	—
Thorax and abdomen	C7–T1/L4–L5*	0.355 LC	0.63	0.37	—	—	—	—
Abdomen and pelvis	T12–L1/greater trochanter*	0.281 PC	0.27	0.73	—	—	—	1.01
Trunk	Greater trochanter/glenohumeral joint*	0.497 M	0.50	0.50	—	—	—	1.03
Trunk head neck	Greater trochanter/glenohumeral joint*	0.578 MC	0.66	0.34 P	0.503	0.830	0.607 M	—
Head, arms, and trunk (HAT)	Greater trochanter/glenohumeral joint*	0.678 MC	0.626	0.374 PC	0.496	0.798	0.621 PC	—
HAT	Greater trochanter/mid rib	0.678	1.142	—	0.903	1.456	—	—

*NOTE: These segments are presented relative to the length between the greater trochanter and the glenohumeral joint.

Source Codes: M, Dempster via Miller and Nelson; *Biomechanics of Sport*, Lea and Febiger, Philadelphia, 1973. P, Dempster via Plagenhoef; *Patterns of Human Motion*, Prentice-Hall, Inc. Englewood Cliffs, NJ, 1971. L, Dempster via Plagenhoef from living subjects; *Patterns of Human Motion*, Prentice-Hall

Winter 2009

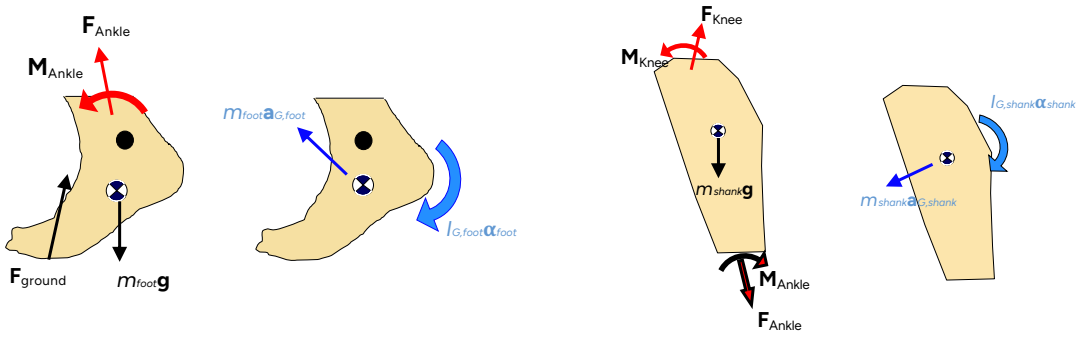
9

(Link segment model, free body diagrams)

Apply the laws of mechanics to each body segment in the model and compute the forces and moments acting at each joint

10

Inverse dynamics: joint moments

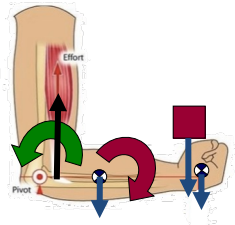


Note: *Segment* angular acceleration
Note: In course book:
- Feet assumed massless and static for ID computations
- Segment angles θ_F, θ_S etc. (θ_1, θ_2) are computed from simple trig. Recall that inverse trigonometry functions have defined ranges. Be sure to apply the correct function, e.g. *atan* vs. *atan2*
- Segment COM $\ddot{y}_G$ and $\ddot{z}_G$ ($\ddot{x}_i$ and $\ddot{y}_i$) are computed analytically, but it's also correct (and easier) to compute them numerically
- Sign convention of moments is opposite; your moment plots should appear upside-down compared to book's.

11

Internal vs. external forces/moments

- External: Moments from forces external to the body
  - Ground reaction forces (GRF)
  - Gravity and inertia
  - External or carried loads
- **Internal: Created by forces within the body to (exactly) counteract external moment**
  - **Muscle tension**
  - **Ligament/soft tissue tension**



(Same same but different)

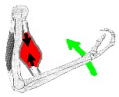
12

Computation of joint power (W or W/kg)

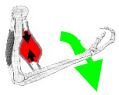
General formula for joint j: $P_j = \mathbf{F}_j \cdot \mathbf{v}_j + \mathbf{M}_j \cdot \boldsymbol{\omega}_j$.
- Power is a *scalar* product of 2 *vectors*. Direction matters.

For a joint with 3 ROM (no translation): $P_j = \mathbf{M}_j \cdot \boldsymbol{\omega}_j$

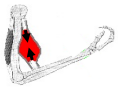
$$P_j = M_{j,x} \cdot \omega_{j,x} + M_{j,y} \cdot \omega_{j,y} + M_{j,z} \cdot \theta_{j,z} \quad \text{(Eq. 1)}$$



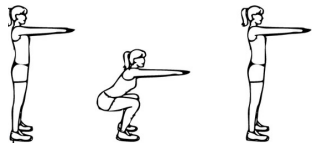
- Positive (same direction) -> Power generation
  Concentric (shortening) muscle action



- Negative (different direction) -> Power absorption
  Eccentric (lengthening) muscle action



- Zero power
  Static (isometric) muscle action

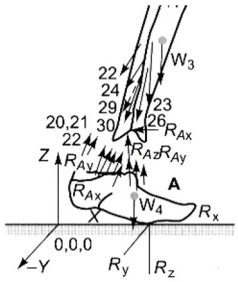


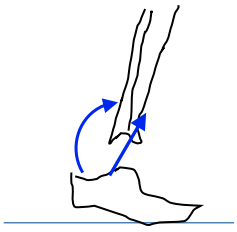
Note:
- In Project 1, you will have to simplify Eq. 1 to: $P_j = M_{j,x} \cdot \omega_{j,x}$ for the sagittal plane
- $\boldsymbol{\omega}_j$ refers to ***joint*** angular velocity, i.e. the time derivative of the *relative joint angle*

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What does inverse dynamics compute?

Net effect of all internal forces and moments of force acting across joints





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What does inverse dynamics *not* compute?

More unknowns than equations!

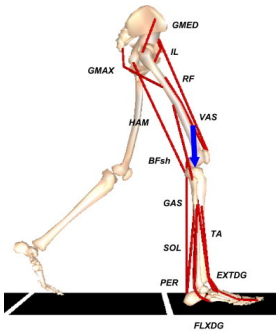
- Individual muscle forces
- Ligament forces
- Joint contact forces

Joint reaction forces ≠ joint contact forces

- Joint contact forces are generally much higher than joint reaction forces

$$\mathbf{F}_C = \mathbf{F}_R + \mathbf{F}_{M,L}$$

(Joint reaction forces alone do not have much intuitive meaning)

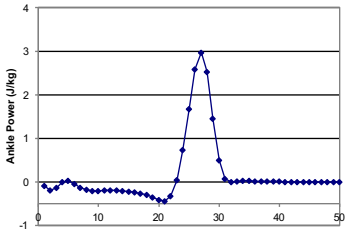


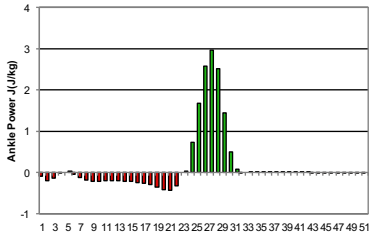
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Joint Work $W_j = \int P_j \, dt$

Net work = change in kinetic energy.

Usually divided into positive work done (the moment adds energy) and negative work done (the moment decreases energy), units J or J/kg.





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Segment and joint angles in 2D vs. 3D

- 2D rotations are simple – one axis of rotation
- 3D rotations are complex: 3 axes of rotation, 3 angles needed.
- Non-commutative; finite rotations are not conserved. Not simple vectors.
  - The numerical values of the angles depend on the sequence of rotations
- Common in biomechanics to use Euler angles (as opposed to e.g. quaternions or helical angles), but gimbal lock can occur.

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3D rigid body rotations

- Segment generally rotated w.r.t. proximal segment's coordinate system
- Cardan series of rotation (*x-y-z: sag-front-trans*) common in biomechanics

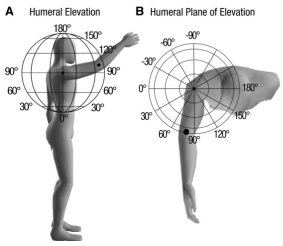
$$[R] = \begin{bmatrix} \cos\beta\cos\gamma & \cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & \sin\gamma\sin\alpha - \cos\gamma\sin\beta\cos\alpha \\ -\sin\gamma\cos\beta & \cos\alpha\cos\gamma - \sin\alpha\sin\beta\sin\gamma & \sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha \\ \sin\beta & -\cos\beta\sin\alpha & \cos\alpha\cos\beta \end{bmatrix}$$

- *z-x-z (Euler series) common in mechanical engineering*
- Suggestion regarding Cardan series: series *z-y-x* agrees with intuitive impression (Ref: R Baker, 2007)

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Segment and joint angles in 2D vs. 3D

- 2D rotations are simple – one axis of rotation
- 3D rotations are comples: 3 axes of rotation, 3 angles needed.
- Non-commutative; finite rotations are not conserved. Not simple vectors.
  - The numerical values of the angles depend on the sequence of rotations
- Common in biomechanics to use Euler angles (as opposed to e.g. quaternions or helical angles) but gimbal lock can occur
- Shoulder: ISB recommendations



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Progress on Project 1

I will upload some examples of segment and joint kinematics and joint moments in a few days. These are to be used to trouble shoot your code/equations.

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